A Governed Closed-Domain Question Answering System for Decision Assistance in Education

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***Abstract*—The recent Artificial intelligence development, proven by the progress in Large Language models (LLMs) and generative systems, has triggered a broad surge in AI adoption in various sectors and domains, especially in the Education Sector. Most of the LLM outcomes are supporting human learning, whether directly through tutoring, explanations, and feedback, or indirectly through improving decision quality, reducing cognitive load, and accelerating access to information that shapes learning-related choices. This is creating a lot of opportunities for the Educational Institutions as the AI powered Question Answering is now expanding to provide more accurate output and guidance. However, the challenge remains with closed domains where we need to keep a certain level of security and privacy. Most of the tasks in the education domain often depend on closed, sensitive, and policy-governed data, where privacy, access control, and answer traceability are essential. In this paper, we propose a governed closed-domain QA framework designed for educational decision assistance across both academic and administrative tasks. The Framework works in three phases. It starts with the query processing step, where we review the user input, then Intelligent Routing & Retrieval, which routes requests to relevant source and finally, retrieval and reinjection, where the responses are produced and constructed into structured outputs.**

Keywords— Education, Closed-Domain Question Answering, Large Language Models, Multilingual NLP, Privacy Governance, Role-Based Access Control, Retrieval-Augmented Generation, Auditability.

I. Introduction

Artificial intelligence (AI) has reached a significant level of maturity. The last two years witnessed a noticeable surge in the use of AI and Question Answering (QA) tools by a broad range of users. It was reported in [1] that, since 2022 and till July 2025, ChatGPT products have been used by 10% of the world's adult population, which is around 800 - 850 million users. The core functions done using these tools can be summarized into the following:

1. Guidance & Tutoring: such as Learning support, ideation, explanations
2. Information Seeking: People, events, recipes, and general knowledge
3. Content Assistance: Email writing, document drafting, summarization, translation

The use of AI has extended beyond education into areas such as customer service, research support, consulting, and information technology workflows [2]. This is widely the case for all LLMs, not just ChatGPT, noting that currently we have a wide variety of LLMs, including Google's Gemini [3] and Anthropic's Claude [4]. The analysis of Claude's interactions shows that the users concentrate heavily on cognitively demanding activities such as problem solving, writing, programming, and troubleshooting [5].

All previously mentioned LLMs are open domains and publicly work for all domains; however, education is a particularly high-impact domain for such systems, as it combines multiple stakeholders and language-intensive processes. The Educational Institutions need to support student needs, such as providing guidance, enabling information retrieval, as well as institution-facing tasks and questions, such as program outcomes and intake of quality, admission process. Also, for the institute, it covers the same points from an administrative point of view. For example, admission and enrollment-related decision processes can be pivotal for program success from the institutional viewpoint [6], [7], while student readiness and support mechanisms are essential from the learner perspective [8]. Accordingly, education offers a strong practical case for QA tools that can interpret user questions using Natural Language processing, locate relevant institutional content, and respond in a way that aligns with institutional expectations.

However, many current education systems target learning assistance, like teacher/student support and classroom usage [9], or use LLMs as a discovery interface for educational resources like a searchable library [10]. These uses are valuable, yet they do not fully address institution-critical data where answers have constraints that need to be followed depending on internal regulations, operational procedures, or sensitive records. In such settings, open-domain generation is insufficient because institutions must ensure that responses originate from the right sources, also respect role-based privileges that are defined based on the domain logic, and finally avoid inappropriate disclosure of protected or personally identifying information constraints commonly associated with closed-domain data access in institutional environments [6].

This challenge becomes more complex in multilingual educational contexts, particularly in Arabic data in institutions. With different languages, the data are distributed across languages, and they might not have one source of truth for the data, with inconsistent terminology and phrasing. Without clear multilingual handling, systems may misclassify the user's question, or retrieve a mismatch, and ideally produce outputs that are difficult to justify or align with the asked question. These limitations are driving us to a closed-domain QA approach in which governance and verification are essential parts of the structure and are embedded within

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the answering workflow rather than applied as an external afterthought.

This paper proposes a private constrained closed-domain question answering framework for decision assistance in education, designed to support institution queries for different users while preserving privacy and enforcing policy constraints. The proposed approach follows three phases/Steps: (1) "Secure Data Preparation": starts with the student queries that will be examined, and then the PII data are extracted and masked, along with the key academic context (such as GPA, credits, major), and then generate and construct a structured version of the question. (2) Intelligent Routing and Retrieval, where the system identifies the task type based on the question and dispatches it to an appropriate pathway. For example, the personal data goes through closed institutional retrieval requests, while the rest of the data goes through the open-data retrieval for general pedagogical guidance, or a study-plan engine for personalized planning, depending on the student's question. (3) Synthesis & Reinjection, at this step, the outputs from the previous step will be validated and evaluated, then sensitive fields are reinjected, and finally, a structured response is generated as the final response and displayed to the user.

This paper is organized as follows: we will start with a full literature review for question answering in general, then specifically for the Education and closed domain. Then move to the proposed framework and finally discuss the conclusion and future work.

II. Literature Review

Question Answering systems have evolved significantly alongside advances in artificial intelligence. The development of Large Language Models (LLMs) has impacted user behavior and work across multiple domains. The education domain is one of the majorly impacted areas, as most of the tasks done using LLMs revolve around learning. In this section, we provide a review of the various approaches, techniques, and architectures for question answering, then review the impact on the education domain.

A. Question-Answering Approaches, Techniques, and Architecture

Question Answering research has advanced massively with the recent AI development. The Question Answering systems started with a rule-based system [11]. The rule-based approach relied on handcrafted linguistic rules, keyword matching, and predefined question patterns, so the effectiveness of these rules depended heavily on the completeness and correctness of encoded rules [12].

These approaches are commonly grouped into 3 types in [13]: Template-based methods that populate predefined answer frames [14]. Syntax-based methods that apply grammatical and logical structures [15]. The semantic-based methods that map language to meaning using ontologies or semantic networks [16].

The rule-based QA performance is good in narrow, controlled domains, as the rules are specific and answer specific questions. However, it typically struggles with language variation, unseen questions, and deeper context and conclusions, which limit scalability and robustness [17]. Accordingly, to improve the flexibility and overcome this limitation, the research started to move toward machine learning (ML) based QA, where subtasks such as question classification, answer selection, and ranking are extracted from annotated corpora rather than specified as rules [18]. Traditional ML uses supervised or semi-supervised classifiers such as Support Vector Machines (SVMs), decision trees, and Conditional Random Fields (CRFs). These classifiers are trained on lexical and shallow semantic features extracted from questions and answers [19].

These feature-driven models improved generalization compared to rule-based systems, yet they remained constrained by feature engineering effort, domain-specific preprocessing, and labeled data requirements, and they often underperformed on deeper semantic understanding and multi-step reasoning limitations that pushed research toward deep representation learning [13].

The next step for QA was deep learning; QA systems learned hierarchical and contextual representations directly from data. The Neural architectures, for example Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and attention-based encoders have created semantic matching improvement between questions and answers, especially for knowledge base QA and neural reading tasks [20]. Surveys show that deep learning improved the contextual extraction and answer accuracy, but at the same time, it increased reliance on large datasets and significant computation, while still facing issues with explainability, robustness, and complex reasoning [21], [22].

Moving from Deep learning, a major turning point happened, which was the Transformer architecture. It started by introducing self-attention for global context modeling and became the key element in modern QA [23]. Building on transformers, pretrained models such as BERT made significant improvements in QA extraction and reading comprehension through fine-tuning. These models outperformed the older version by improving their extraction through using large datasets such as SQuAD [24]. The transformer-based systems then evolved toward generative QA and open-domain settings, which included retrieval-augmented designs that enhanced factual grounding by merging retrieval with generation [25].

Recently, the emergence of Large Language Models (LLMs) changed QA approaches drastically towards a general-purpose reasoning that uses natural language generation. It usually generates the answers using pretraining, which is represented by the GPT-style models. The transformer-based retrieval-generation systems (RAG [25]) and LLMs (GPT-4 [26], Claude 3 [27], Gemini [3]) enable broad reasoning and generation and demonstrate strong zero-shot and few-shot QA as model and data scale increased [28]. These LLMS can be fine-tuned using specific datasets, which create what is called instruction-tuned models and reinforcement learning from human feedback (RLHF). These are used in InstructGPT [29], which results in more controlled and reliable answers.

Newer models such as GPT-4 [28], Claude 3[29], and Gemini [3] extended the QA progress to a stronger reasoning, multilingual interaction, and long-context or multimodal capabilities. But, because LLM knowledge is bound by

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pretraining, these systems can still produce incorrect or outdated outputs.

This ideally drives into more hybrid designs which use Retrieval-Augmented Generation (RAG) [25] to mitigate hallucination and improve grounding, especially for education contexts. As per [30], the educational literature emphasizes RAG as especially relevant where correctness, transparency, and alignment with approved content are required, supporting learning and institutional use cases with better reliability and traceability than standalone LLMs. Fig. shows the RAG workflow in education, considering both students and teachers as the primary users of the tool.

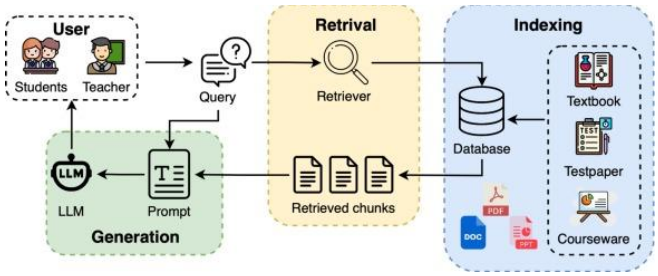


Fig. 1. The workflow of RAG [30].

Finally, hybrid QA combines symbolic and neural components, usually using knowledge graphs (Ontology) to support structured reasoning while keeping the balance [31]. The improvement happens with embedding and integrating structured knowledge based on the available data sets, which ideally improves multi-hop reasoning and context-aware answering [32]. In [33], the recent hybrid LLMs showed improved control using validation rules, schemas, and ontologies, which highly reduced the hallucination and improved reliability. This is extremely useful in safety-critical domains such as education.

Along with the approaches, changes, and development, the techniques used in each approach are also developed. It goes 3 sets of techniques: the question analysis techniques, Information retrieval, and answer extraction and generation. These different techniques aim to strengthen the query of understanding, question classification, and expected answer types. The classical Machine learning methods, such as Support Vector Machine (SVMs) and logistic regression, are still used for question classification as it works with constrained and even with low-resource settings [34]. They matter the most with the limited training data, which reduces the feasibility of more complex models [19].

The query analysis techniques aim to transform the raw natural language and understand the user's intent using lexical, syntactic, and semantic processing. It includes query expansion and handling out-of-vocabulary terms (OVVs), and using contextual embeddings, which, at the end, improve precision and F1, though may affect ranking stability [35]. The extraction is done using Preprocessing such as segmentation, keyword extraction, and answer-type detection, which is also widely recognized as a critical layer in QA architecture pipelines [36].

Once keywords are extracted, our focus will shift to information retrieval, which remains central for QA, as this is the stage where we find the evidence to then extract the answer. The retrieval methods start with the sparse retrieval, such as BM25, which is Term Frequency-Inverse Document Frequency (TF-IDF) weighting and ranking functions for evidence and document estimation [37]. Then, moving from the BM25 used for efficiency and strong lexical matching to the K-Nearest Neighbors (KNN), measures the similarity between query vectors and indexed documents [30]. The Second retrieval method is the Dense retrieval, which enables semantic search by mapping queries and documents into shared spaces, with methods such as DPR [38], ColBERTv2 [39], and Contriever [40]. It shows a wide improvement in the retrieval's quality and generalization. Knowing the techniques, modern tools mostly use a hybrid approach between sparse recall and dense ranking.

After retrieval, systems start the extraction process with different extraction techniques, such as span selection using transformer encoders and attention, or generative techniques such as sequence-to-sequence and autoregressive generation models [13]. More recently, RAG has become a dominant technique for generation, using strategies such as Fusion-in-Decoder (FiD), re-ranking, and multi-document aggregation [41]. It is improved further using prompt engineering, domain fine-tuning, and adaptive retrieval, especially relevant for educational settings [42].

Considering all the techniques and approaches that are used to create question answering, a lot of architectures have been produced to orchestrate the work between all the different tools and techniques. The typical architecture for QA is implemented as a multi-stage architecture. This architecture starts with a query and then transforms it into an answer through modular components. The pipeline begins with question processing, which includes tokenization, syntactic parsing, question classification, and semantic interpretation to extract the user's intent and answer type. This step is crucial in the retrieval strategy and response formulation [43]. The second stage is information retrieval, where the system searches for the relevant evidence and documents from available corpora, datasets, knowledge bases, or hybrid repositories. Earlier architectures relied on keyword methods such as TF-IDF and BM25, then moved to the dense retrieval based on transformer embeddings or dual-encoder designs, and finally, the hybrid approach as mentioned before [38].

The last stage is answering processing, where systems either extract spans (extractive QA) or generate natural-language responses using generative QA and LLMS based on the retrieved context. It also includes retrieval-augmented generation models to ensure that the mode outputs are supported by the provided information and knowledge page, or what is called improving the factual grounding [44].

Many architectures then apply answer selection and re-ranking using scoring functions or neural rankers to ensure relevance and coherence. In high-reliability domains, post-processing layers may include rule-based validation, factual consistency checks, and justification mechanisms to increase trustworthiness. Even when end-to-end LLMs appear to compress these stages into a single model, modern QA systems still frequently preserve retrieval augmentation and validation layers to maintain grounded answers and support domain-specific requirements, including educational use cases [38], [44].

Many QA architectures select the best answer, rank the candidate's answers, and reorder using scoring methods and ranking models to show relevant and clear results. In a closed private domain setting, they add an extra layer after

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generation, such as rule-based validation, fact checks, and short explanations to make the answer more reliable. Even if the final LLM seems to do everything in one step, most QA systems are using layers and steps for retrieval and validation layers to keep answers grounded and meet domain needs, including education [38], [44].

B. Educational QA Systems: Datasets and Closed-Domain Constraints

The Educational Domain is one of the most influential domains; it has been impacted by the developments in LLMs. Both students and teachers are using the open-access LLMs for learning and support [1]. In [30], the author surveyed 47 RAG based questions answering systems used in education, as summarized in Figure 2. The usage was summarized into three main categories: supporting the education ecosystem, assessments, giving feedback, and students as QA systems and chatbots.

A fourth category can be added based on the work done in [6], which is adding the institution's decision makers and administrators to the users we can support by addressing the decision process needed for student recruitment for academic programs. This was expanded further in [7] by introducing bilingual support and multi-language data sources to enable cross-source and cross-language answering.

Education Datasets successfully positioned itself among other datasets, such as SQuAD [45], TriviaQA [46], HOTPOTQA [47] and MS MARCO datasets. All the previously mentioned datasets are public datasets that contributed widely to the open domain QA system.

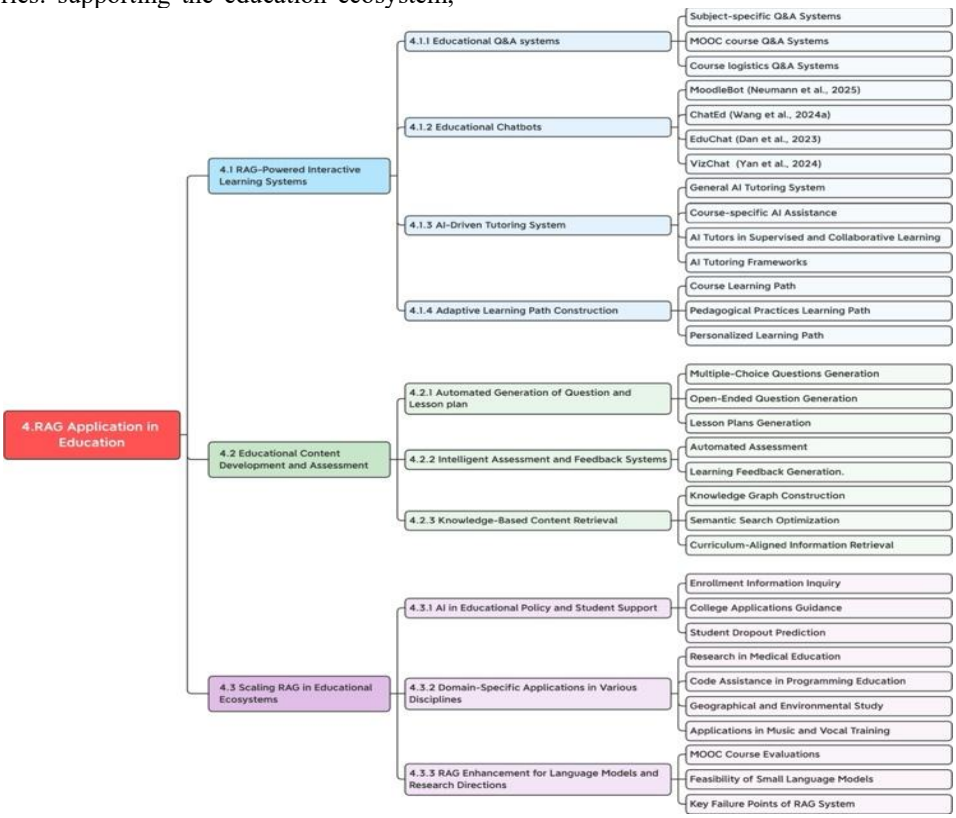


Fig. 2. Summary of RAG Application in Education [30].

However, when it comes to the education domain as a closed domain, the case is different, as each institute uses its own data.

As the Educational domain is closed, it usually depends on the institute's data sources. When we reviewed the QA Systems in education, we found that all the QAs depend on the closed domain datasets. All QA are dependent on the internal datasets, for example, the education FAQ-style chatbot, which was using universities' datasets to provide answers [8]. Same for [6],[7], it was using internal data sources to support decisions, while giving the option to create an internal data repository for the institution. Another QA for selecting a college major while making sure the responses are not biased or prejudiced towards a specific person. It states that the datasets play a significant role in the response being biased or not [48]. In [49], another QA to support students in academic advising was using human advisors to train the queries and used it as a reference to train.

The dependency on confidential/personal data, according to the domain, sometimes limits the use of AI and the quality of the results. In [50] and evaluation was made for the quality of QA in Education, listing benefits and challenges. A main challenge was towards Privacy & Security, also the Bias happening due to the data size, and lastly the huge cost for implementation. In another study [51] assessing the use of chatbots in academic advising, it stated that data availability is one of the major limitations, especially with the Arabic language. However, it was proven useful for the students. Another QA tried to merge the Academic advising with a personalized plan for the user based on their progress and degrees [52].

Several QA systems were created to support education in Arabic. An Arabic chatbot to help students in college admission [53]. A Review was conducted on the Arabic language chatbots and the technologies used in them; most of the chatbots use retrieval-based approach with NLP and deep learning [54]. Ideally, that happens due to the domain and the

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closed datasets. In [55], the importance of machine learning and Natural Language processing techniques for Arabic datasets is highlighted. The difficulty of the language and dialects of Arabic and different options to overcome the scarcity of Arabic datasets through translation is highlighted in [6]. Another tool is built to help non-native students learn Arabic using Visual Question Answering (VQA). It creates quizzes with AI models and was tested on 1,266 with human feedback. [56]

Accordingly, privacy is causing limitations in terms of the datasets we are using and causing Biase especially if the dataset size is small. Although (RAG) reduces these risks by limiting the responses from specific approved repositories. However, with the new LLMs and open tools, it started to increase the privacy gap and sensitive data sharing [50]. Recent research proposed a privacy layer, "LLM Access Shield," which targets masking sensitive data before sending it to the public LLMs. This enforces privacy policies and reduces the leakage risk while making sure the tasks are done with the context [57].

III. Proposed Framework and Methodology

This research proposed a framework that targets privacy and security. It is stated as a governed closed-domain question answering (QA) framework for decision assistance in education, which is designed to operate under institutional constraints related to privacy, access control, and auditability. The framework targets both academic and administrative areas where confidential/personal data should be constrained and used anonymously, and the answers must align and be reviewed based on the approved institutional knowledge rather than unconstrained open-domain sources. It targets assistance in scenarios such as advising, academic path, Score evaluation, curriculum clarification, and study planning. To apply this, the proposed design adopts a controlled, multi-stage pipeline that minimizes exposure of sensitive data, restricts retrieval scope based on task needs, and enforces validation before returning responses.

A. Framework Overview and Design Goals

The proposed framework is designed to support privacy using privacy-aware question answering. It is combining approved closed institutional sources with public data; at the same time, it is keeping a clear control over what can be used in each case. Instead of treating all queries the same, the framework applies the routing rules that decide which parts of the request can be handled using public information through LLMs, and which parts must be answered using internal institutional sources and processes.

This design is generated based on the privacy constraints and user needs in educational QA. Most of the questions are classified based on the tasks, such as admissions, curriculum, student services, and career evaluation & personalized study Plan. So, routing the query to the right flow improves relevance and avoids unnecessary retrieval.

The framework aims to merge the best of both approaches; we can use the fast-developed LLMs freely with the allowed public and open data in the educational domain. While keeping all confidential data as a second layer and applying internal rules and extraction based on the internal records and confidential/Personal data sources. So, we can list the data sources for this system as public knowledge (Online sources), institutional data sources and repositories, domain rules, and constraints.

Thus, the framework follows a few clear sets of goals:

1. **Data source Controlling,** as the system evaluates and controls what to share publicly and what not to share intentionally through routing.
2. **Privacy by design,** as it limits the exposure of personal or sensitive data and enforces masking when needed.
3. **Task routing,** send each query through the most suitable path.
4. **Auditability and Logs**, by keeping a clear log and evidence of what was accessed and how the answer was formed.

B. Proposed Framework Methodology and Workflow

The main idea is to separate what can be processed using public data using the LLM reasoning from what must be processed internally using approved institutional sources to keep privacy and governance controlled across the architecture. The framework in Figure 3 explains the high-level of the phases and steps to achieve this framework.

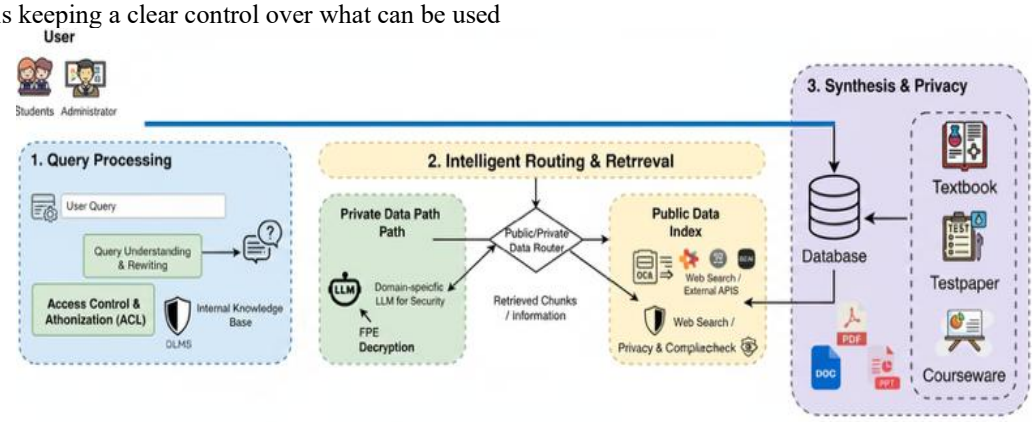


Fig. 3. Proposed Architectural Framework for a Governed Closed-Educational Domain.

As shown in Figure 3, the framework starts with the Query processing phase, where the user provides a question in either Arabic or English. The system processes the question through parsing and NLP extraction to identify keywords. The system identifies user data within the question (Personal and Education data), then moves to the intelligent routing & Retrieval phase.

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In this phase, the first step is to query the confidential/personal data in the institution database to verify the accuracy of the provided data and the user's authority to access this information. Once verified, the system detects the question type using the keywords matching across four main task types: Admissions, Curriculum, Student Services, and Career Evaluation & Personalized Plan. It then applies the following routing:

1. Admission: Matches keywords to specific program requirements. The system redirects the user to the internal dataset for Admission and Program requirements.
2. Curriculum & Study: Handles general academic queries; it redirects the user to a Public LLM.
3. Student Services: If the question keywords match, the system redirects the user to the internal LLM to answer institution-specific questions (e.g., campus facilities).
4. Career Evaluation & Personalized Plan: Merges Internal LLM data with a new query for broader context which is then passed to the Public LLM for further evaluation and cases.

Once routing is complete and results are retrieved, the final Synthesis & Privacy phase begins by producing a clear, structured output such as an academic roadmap or a verified policy answer and delivers the final response to the student.

IV. Implementation

Building on the proposed framework, this section presents the Python implementation of a RAG pipeline. We use CAMeL Tools [58] for Arabic NLP analysis and spaCy [59] for English NLP, and we use GPT [28] to handle public LLM tasks. It is implemented in three stages: Initial Data Processing & Query, Intelligent Routing & Mapping, and Synthesis & Final Review

In Figure 4, we present the workflow of the proposed framework using an example that illustrates how a student query moves from intake to a final, validated response. The framework proceeds through three phases. The following sections provide details for each step, along with a student interaction use-case example.

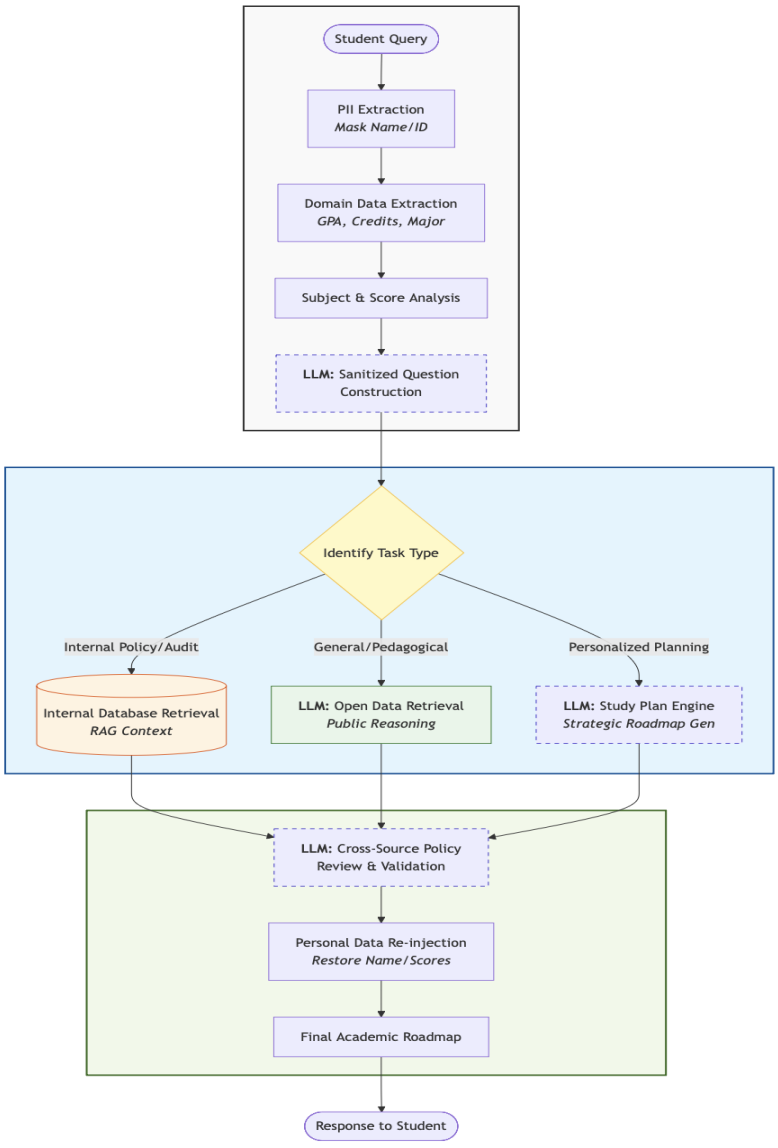


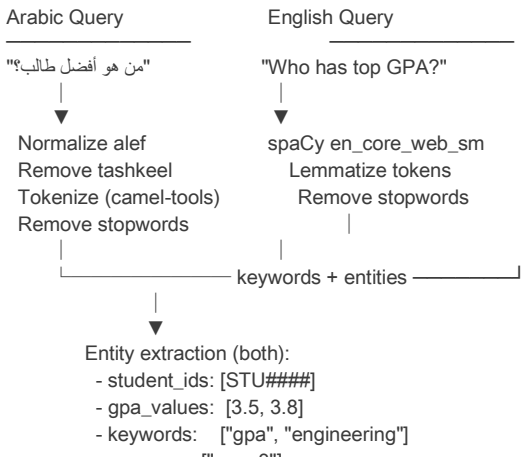
Fig. 4. Application of Student Query on the Framework

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A. Initial Data Processing & Query

The flow starts with Query Processing; The workflow starts with the student's query. The student query goes through NLP parsing engine based on the users input language Arabic or English. System extracts personal identifiable information (PII), such as name, age, phone number, email, etc., and the education information such as Student ID, grades, GPA, major, and Credits to limit exposure of sensitive information. Once processing is done, the system generates the keywords and classifies the data "PII Data, and Academic Data."

Once Question is received, the system starts the NLP processing by detecting the language. In case of Arabic question, starts by normalizing alef variants (ا → آ أ إ), remove diacritics (تشكيل), tokenize with camel-tools then remove the stop words. In case of English, it starts with spaCy lemmatization, then remove stop words. Once done, entity extraction such as student name, IDs, GPA values (e.g., "3.5") and Keywords.



Example: A Student asks the following question: "I am struggling with Calculus (score: 62) but doing great in Physics (score: 95). How can I improve my overall GPA?" Question Analysis starts, it is being analyzed as mentioned before, and system extracts and provides the keywords.


```
{
  "PII": {
    "name": "Sarah Miller",
    "id": "8829"
  },
  "Domain": {
    "gpa": 3.1,
    "subject_scores": {
      "Calculus": 62,
      "Physics": 95
    },
    "major": "الهندسة"
  },
  "Keywords": [
    "GPA improvement",
    "Calculus support",
    "study plan"
  ]
```


Before moving to the next step system, review the user's access and validate the mentioned data in the user's input (Validate and confirm if the data is correct).

B. Intelligent Routing & Mapping

At this stage, we take the output from stage 1 as input. System starts the keywords mapping with the education task types "Admissions, Curriculum, Student Services, and Career Evaluation & Personalized Plan" to decide the appropriate processing path.

If the query is related to policy or auditing and needs institutional evidence, it is routed to internal retrieval over approved repositories to build the controlled context using RAG. If the query is general learning support, it can rely on public data with LLM reasoning.

Example: As a next step based on the keywords system detect the track is "Career Evaluation & Personalized Plan" and accordingly route the process as follows:

1. **Internal RAG System and Data:** the system starts internal search in both Arabic & English for 'Academic Policies' and 'Support Services' for [GPA improvement], [Engineering Tutoring Schedules], and [GPA Calculation Rules for Engineering Majors]. Filter for relevance to Grade < 65%
2. **Prompt to Public LLM:** Based on the keywords " GPA Improvement" and "Calculus Support," the system synthesizes the public prompt: "Struggling with Calculus." Propose an improvement plan for overall GPA."

Once completed, the system receives the outputs of these flows and passes them as input to the next and final stage.

Synthesis & Final Review

In this final phase, the system reviews the previously erated results against the access policy and the allowed ı scope before returning them. Once reviewed, the system ıtes and synthesizes the answer with the personal data after cting the personal data only when needed and permitted. The framework then produces a clear, structured output, such as an academic roadmap or a verified policy answer, and delivers the final response to the student.

Once data is returned, the system evaluates the response and starts PII and domain data embedding inside the content. They provide the final delivery message for the user, stating actions and a plan for the user to enhance their performance. Then, at the end, a log of the task is generated with the Task ID, PII Object, Domain Objects, Queries, and answers.

V. Evaluation

We evaluated the implementation using Retrieval Augmented Generation Assessment (RAGAS) [60]. It was deployed over an anonymized institutional student-records dataset that (e.g., GPA, academic status, advisor, city, Academic track,
scholarship, and attendance). The identifiers were replaced with codes (e.g., STU0001). The benchmark includes 36 deterministic Arabic test queries over 14 categories (e.g., top-GPA, lookup by ID, probation/status filtering, advisor and city-based queries, GPA thresholds, scholarship, attendance). The evaluation was done for the following aspects: Retrieval effectiveness, Answer quality, and End to End latency.

Overall retrieval performance was modest (Precision@5 = 0.206, Recall@5 = 0.380, F1@5 = 0.243). MRR@5 = 0.517

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value suggests relevant results frequently appear within the top three candidates. Hit@1 was 0.417. The analysis exposes a semantic structural gap when passing semantically rich queries with structured lockup which indicates the need for hybrid retrieval (metadata/SQL style filters combined with the semantic search).

Answer quality is also impacted by the Arabic usage; answer relevancy (0.404) and faithfulness (0.592) were low due to malformed Arabic output. However, the context of recall was comparatively high (0.549), showing that relevant evidence is often retrieved. Overall, the evaluation validates the framework's potential for multilingual institutional QA (notably evidence coverage in retrieved context)

End to end latency for the question coming from the internal RAG to the mean (1,083ms), while P95 (1,935ms). However, this does not represent the full End to end latency as the framework is dependent on external sources for some questions which will increase the latency.

Overall, the framework satisfies the needs and fills the addressed gap in the education domain. Gather all organizational data into one source of truth and give access to the domain users as per their needs and constraints also limit the data leakage that could happen from using public LLMs.

VI. Conclusion & Future Work

In this paper, we proposed a closed domain question answering system that supports education constraints to make sure privacy is preserved for the users. Based on the domain role and constraints, we provided a well-structured and robust tool for the institute to support the different users and people in the organization. While privacy check might cause latency for the output; however, it ensures that the response is following the regulation, such as General Data Protection Regulation (GDPR), which is required in the European Union (EU), and Family Educational Rights and Privacy Act (FERPA), which is required in the USA.

Despite the strengths, there are some limitations that we need to improve. It is required to enhance latency and perform end-to-end testing for the model and outputs considering the external factor. Reduce the semantic gap that happens due to removing, then injecting the personal data into the desired answer. Work on an ongoing conversation handling especially with complex queries that have multiple intents. We aim to improve daily usability in a real university environment by working on pilot deployment inside it.

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